



Dynamic Dictionary Learning for Remote Sensing Image Segmentation

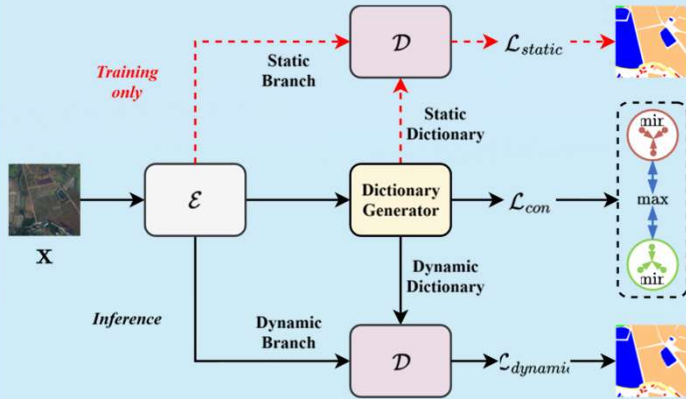
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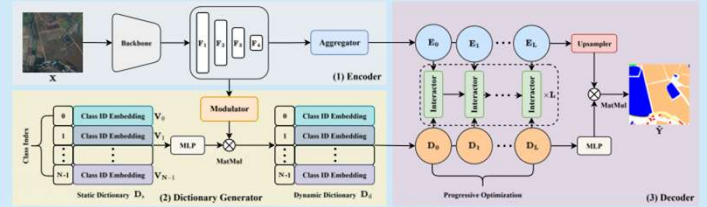
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1 Motivation & Pipeline

Remote sensing image segmentation is challenged by intra-class variation and inter-class similarity. D2LS learns adaptive and discriminative class embeddings with a dynamic dictionary learning framework.



2 Network Architecture



Method Highlights

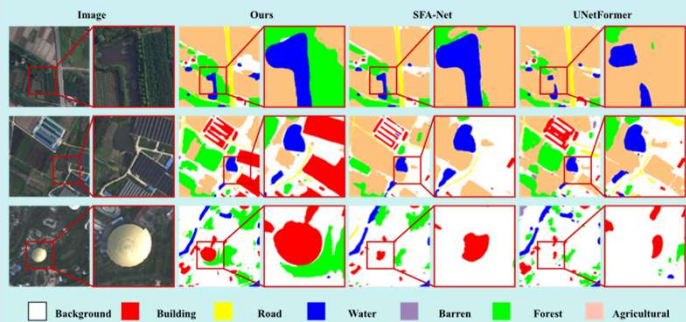
- Dynamic and class-aware dictionaries with contrastive learning for stronger class separability.
- A modulator converts static dictionaries into input-specific dynamic dictionaries.
- Multi-stage alternating cross-attention progressively refines features and embeddings.

3 Experimental Results

Quantitative Results

LoveDA (mIoU, %)		UAVID (mIoU, %)		Potsdam (mF1, %)		Vaihingen (mF1, %)	
Method	mIoU	Method	mIoU	Method	mF1	Method	mF1
AerialFormer	54.1	AerialFormer	54.1	AerialFormer	67.8	AerialFormer	67.8
SFA-Net	54.9	SFA-Net	54.9	SFA-Net	70.4	SFA-Net	70.4
D2LS (Ours)	55.3	D2LS (Ours)	70.9	D2LS (Ours)	94.7	D2LS (Ours)	91.9

Qualitative Results on the LoveDA dataset



4 Ablation Studies

Explicit Class Modeling

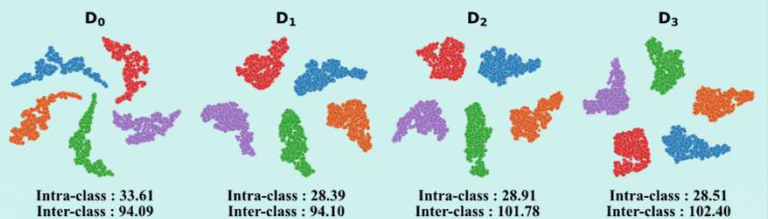
Dictionary Generator	LoveDA	Vaihingen	Grass
×	54.81	84.40	50.70
✓	55.27	85.27	51.96

Static Dictionary vs. Dynamic Dictionary

Loss Function	LoveDA	Vaihingen	Grass
$\mathcal{L}_{static}$	54.80	85.10	50.96
$\mathcal{L}_{dynamic}$	54.93	85.17	51.38
$\mathcal{L}_{static} + \mathcal{L}_{dynamic}$	54.95	85.17	51.53
$\mathcal{L}_{total}$	55.27	85.27	51.96

5 Embedding Visualization (t-SNE)

Our code is publicly available!



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